EXPERIMENT 9: IMPLEMENTATION OF BIT STUFFING MECHANISM USING C

PROGRAM :

```
#include <stdio.h>
#include <string.h>
int main(void) {
    char data[200], stuffed[420];
    int i, j = 0, count = 0;
    printf("Enter data bits (0s and 1s only): ");
    scanf("%199s", data);
    strcpy(stuffed, "01111110");
    j = strlen(stuffed);
    for (i = 0; data[i] != '\0'; i++) {
        if (data[i] != '0' && data[i] != '1') {
            printf("Invalid input: enter only 0 and 1.\n");
            return 1;
        }
        stuffed[j++] = data[i];
        if (data[i] == '1') {
            count++;
            if (count == 5) {
                stuffed[j++] = '0';
                count = 0;
            }
        } else {
```

```
            count = 0;
        }
    }
    strcpy(stuffed + j, "01111110");
    printf("Original data : %s\n", data);
    printf("Stuffed frame : %s\n", stuffed);
    return 0;
}
```

OUTPUT :

